

Spatial Heterogeneity in COVID-19 Response: Leveraging Regression Models to Explore Socioeconomic and Demographic Disparities in New York City

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I. Overview and Student Role

Project Overview:

This study analyzed United Hospital Fund neighborhood-level population data to assess socioeconomic and demographic disparities across regions in New York City and their influence on health crisis decision-making during the COVID-19 pandemic. By integrating census data and decision-making indices, this study applied multiple linear regression (MLR) and geographically weighted regression (GWR) models to quantify spatial variations in pandemic responses. The GWR framework was further leveraged as an effective method for predicting response scores, preserving spatial heterogeneity. Results revealed significant interactions between socioeconomic factors (e.g., income, education) and health-related response behaviors, demonstrating how localized disparities influenced decision-making indices. These insights underscore the importance of spatially adaptive models in public health planning and might offer actionable strategies for improving equity in future crisis responses.

Student Contribution:

- Collected, cleaned, filtered, and standardized data.
- Conducted exploratory data analysis (EDA) on neighborhood-level census data and decision-making indices in response to scores.
- Applied regression models to predict pandemic response scores across UHF neighborhoods and visualized spatial patterns.
- Compared model performance to identify optimal approaches.

II. Background

Research Context:

The COVID-19 pandemic underscored the critical role of socioeconomic and demographic factors in shaping decision-making and responses to health crises. This study analyzes neighborhood-level data from New York City’s United Hospital Fund regions to investigate these dynamics, focusing on three behavioral metrics: agency score, temporal discounting score, and loss aversion score. By evaluating these scores alongside socioeconomic and demographic data, the study identifies how structural factors interacted to shape pandemic behaviors, offering actionable insights for tailoring equitable public health strategies.

Key Concepts:

- NYC United Hospital Fund 42 Neighborhoods: Consists of 42 contiguous zip code areas, designed to align closely with New York City’s community planning districts; serves as a geographic framework for analyzing demographic data at a localized level.
- Agency Score [1]: Assesses individuals’ capacity to autonomously form intentions and act on them, reflecting empowerment in health-related decision-making.
- Temporal Discounting Score [2]: Measures the propensity to prioritize immediate rewards over future benefits, influencing compliance with long-term preventive health measures.
- Loss Aversion Score [3]: Evaluates the tendency to avoid losses rather than pursue gains, shaping risk-avoidance behaviors during health crises.

III. Methods

Data Sources:

- Scores Data: A survey is designed to measure scores using Qualtrics. Temporal discounting involved dynamic financial decisions where participants chose between immediate and delayed monetary rewards. Loss aversion was evaluated through preferences for certain or probabilistic gains and losses. Agency was assessed using a new developed scale [4] measuring individuals’ perceived ability to engage with public safety measures.
- Census Data: The census data is sourced from the U.S. Census Bureau’s official platform, data.census.gov, which provides access to a wide range of demographic, economic, and social statistics for the United States.

- **Map Data:** The map data is from the New York City Department of Health and Mental Hygiene website. It provides a comprehensive collection of geographic information system (GIS) data files and maps related to public health in NYC, all geocoded for spatial analysis.

Data Processing:

- **Scores Data:** The processing steps included merging multiple datasets by participant ID, standardizing scores, and mapping zip codes to UHF regions. The data was summarized by UHF region and filtered to include only regions with at least 15 participants, resulting in a final dataset covering 18 UHF regions with 1,344 observations. For each region, score values were calculated by the median of all participants' scores.
- **Census Data:** The processing steps involved downloading and merging U.S. Census Bureau ACS data at the ZCTA level, mapping ZCTAs to UHF regions, and aggregating demographic and socioeconomic variables (e.g., education, employment, household income, and age distributions) by UHF region. The final dataset covers 42 UHF regions.

Analytical Methods:

The study employs two regression approaches: Multiple Linear Regression and Geographically Weighted Regression. Both models predict the median scores using variables including insurance coverage, bachelor's degree attainment, household income, household number, vehicle ownership, weighted age, and employment.

Specifically, the study uses the percentage or average values of these predictors for each region to calculate the median score, accounting for spatial heterogeneity. For instance, a value of bachelor = 15.9 in UHF 101 indicates that 15.9% of the population in that region holds a bachelor's degree. Additionally, the weighted age is calculated by multiplying the percentage of each age group by its corresponding age.

Multiple Linear Regression

$$\begin{aligned} \text{medianscore}_i = & \beta_0 + \beta_1 \cdot \text{no_insurance}_i + \beta_2 \cdot \text{bachelor}_i + \beta_3 \cdot \text{household_income}_i \\ & + \beta_4 \cdot \text{household_num}_i + \beta_5 \cdot \text{no_vehicles}_i + \beta_6 \cdot \text{weighted_age}_i \\ & + \beta_7 \cdot \text{employment}_i + \varepsilon_i \end{aligned}$$

Geographically Weighted Regression

$$\begin{aligned} \text{medianscore}_i = & \beta_0(u_i, v_i) + \beta_1(u_i, v_i) \cdot \text{no_insurance}_i + \beta_2(u_i, v_i) \cdot \text{bachelor}_i \\ & + \beta_3(u_i, v_i) \cdot \text{household_income}_i + \beta_4(u_i, v_i) \cdot \text{household_num}_i \\ & + \beta_5(u_i, v_i) \cdot \text{no_vehicles}_i + \beta_6(u_i, v_i) \cdot \text{weighted_age}_i + \beta_7(u_i, v_i) \\ & \cdot \text{employment}_i + \varepsilon_i \end{aligned}$$

where:

- (u_i, v_i) are the spatial coordinates of the i -th observation,
- $\beta_0(u_i, v_i), \beta_1(u_i, v_i), \dots, \beta_7(u_i, v_i)$ are the spatially varying coefficients,
- ε_i is the error term for the i -th observation.

IV. Results

Exploratory Data Analysis:

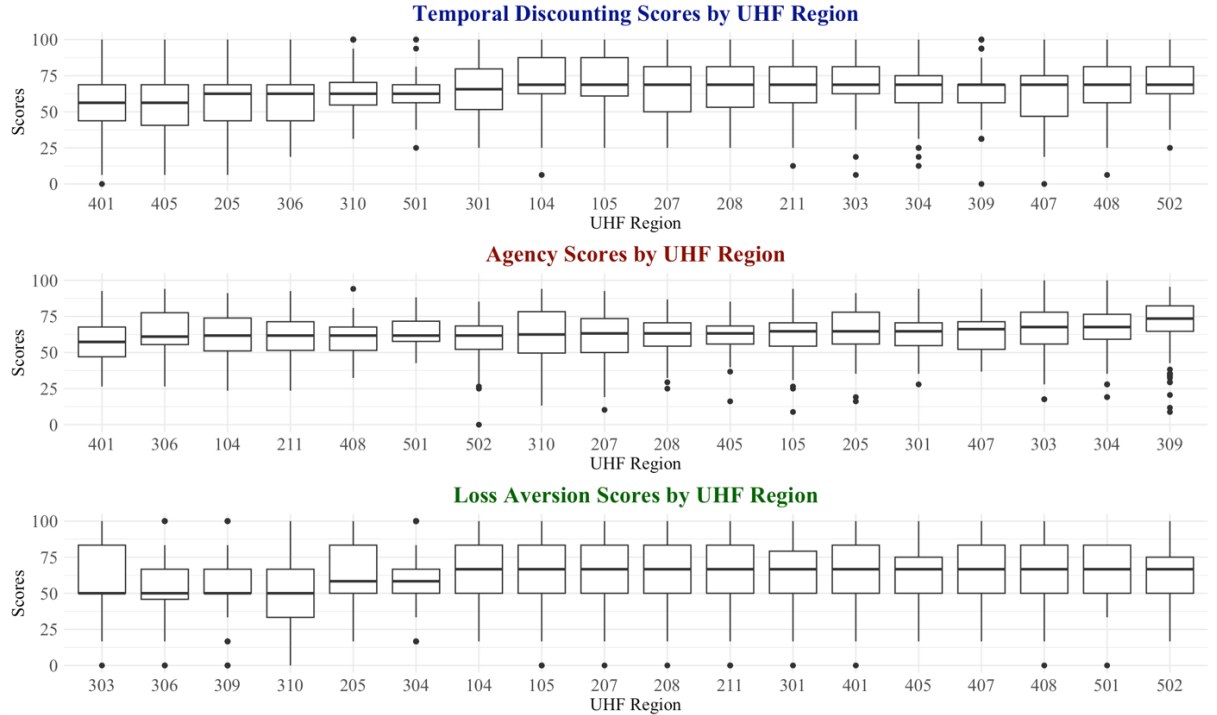


Figure 1. Boxplot of Scores by UHF Region

The analysis revealed that UHF 401 (Long Island City: Astoria) exhibited the lowest median temporal discounting scores. This finding aligns with intuitive expectations, as socioeconomic data indicate that Astoria has an above-average household income. These factors may make individuals more inclined to delay immediate gratification in favor of long-term benefits. In contrast, UHF 309 (Union Square: Lower East Side) exhibited the highest median agency score, which may be attributed to the area's well-developed public transportation system and the accessibility of convenience stores. These factors enhance individuals' ability to make independent choices and exercise greater autonomy in their daily activities.

Model Performance:

Table 1. Results of Multiple Linear Regression

	td: coefficients (se)	ag: coefficients (se)	la: coefficients (se)
intercept	118.25*** (26.07)	108.49*** (16.79)	77.37 (37.48)
health insurance	3.34** (0.88)	-0.01 (0.57)	1.92 (1.27)
bachelor	-0.64 (0.56)	1.30** (0.36)	-1.11 (0.80)
household income	-0.01 (0.00)	-0.00 (0.00)	-0.01 (0.00)
household number	-1.49** (0.47)	-1.13** (0.30)	-0.22 (0.67)
vehicles	-0.68** (0.18)	-0.42** (0.12)	0.32 (0.26)
weighted age	0.15 (0.41)	0.47 (0.27)	-0.74 (0.59)
employment	0.41 (0.46)	-1.11** (0.29)	1.38 (0.66)

- Adjusted R-squared values: td model: 0.6095, ag model: 0.7462, la model: 0.6998.
- Abbreviations: td represents median temporal discounting scores, ag represents median agency scores, and la represents median loss aversion scores.

The findings indicate that the average household number and vehicle ownership in a region exert a negative influence on both median temporal discounting and median agency scores. This suggests a regional propensity for favoring long-term benefits and a potential reliance on policy-driven healthcare decisions. Employment status has a significant positive effect on the median loss aversion score: those who are employed are likely to have greater financial and job security and thus make more prudent decisions to avoid losses.

Table 2. Results of Geographically Weighted Regression

	td: coefficients (se)	ag: coefficients (se)	la: coefficients (se)
intercept	113.38*** (26.07)	99.83*** (16.79)	76.51*** (37.48)
health insurance	2.41** (0.88)	-0.18 (0.57)	3.34** (1.27)
bachelor	-0.37 (0.56)	1.25*** (0.36)	-1.86** (0.80)
household income	-0.00 (0.00)	-0.00 (0.00)	-0.01 (0.00)
household number	-0.95** (0.47)	-1.03*** (0.30)	-0.51 (0.67)
vehicles	-0.39** (0.18)	-0.34** (0.12)	0.10 (0.26)
weighted age	0.09 (0.41)	0.56*** (0.27)	-0.84 (0.59)
employment	0.00 (0.46)	-1.06*** (0.29)	2.12*** (0.66)

- Quasi-global R-squared values: td model: 0.9968, ag model: 0.7916, la model: 0.8784.
- Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

In comparison to the MLR model, the GWR model identified a greater number of significant predictors. This finding underscores the necessity of accounting for spatial variation when understanding regional preference reactions to healthcare policies. The GWR model's high quasi-global R-squared values further substantiate its capacity to capture spatial patterns.

The study then employed a Spatial ANOVA [5] to compare GWR with MLR models. GWR demonstrated a significant improvement in prediction for the temporal discounting score (SS improvement = 124.28, $F = 142.62$, $p < 0.001$) and loss aversion score (SS improvement = 11.79, $F = 3.72$, $p = 0.001$), but not for the agency score (SS improvement = 1.30, $F = 1.44$, $p = 0.193$). This finding suggests that temporal discounting and loss aversion exhibited stronger spatial heterogeneity, likely due to region-specific factors such as economic conditions or policy environments. In contrast, agency scores may reflect more universal psychological traits or behaviors that are less influenced by geographic context.

Visualization:

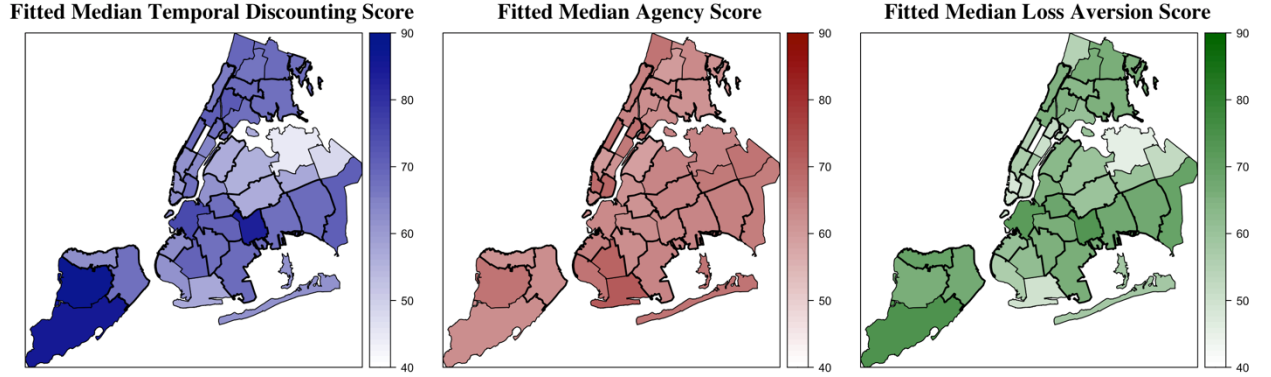


Figure 2. Spatial Distribution of Fitted Scores Across NYC Neighborhoods

The three maps illustrate the spatial distribution of fitted median scores for temporal discounting, agency, and loss aversion scores across NYC neighborhood regions. Specifically, darker-colored areas indicate higher fitted scores, while lighter areas represent lower scores. The black-outlined regions contain the original median score data, while the rest of the map shows predictions generated using GWR models based on socio-economic and demographic data. This visualization provides an intuitive way to understand regional differences in decision-making tendencies, making the findings accessible to a broad audience.

V. Conclusion/Discussions

Key Findings:

During the COVID-19 pandemic, New York City residents demonstrated divergent decision-making patterns, a phenomenon attributable to socioeconomic and demographic factors. The GWR model exhibited superiority over the MLR model's capacity to capture these spatial heterogeneities. Wealthier districts exhibited lower temporal discounting scores, signifying that their residents exhibited a greater propensity to prioritize long-term benefits. Furthermore, employment status exhibited a robust correlation with loss aversion, with employed individuals demonstrating a heightened tendency towards risk aversion. However, agency scores, which reflect decision-making autonomy, exhibited minimal variation across locations, indicating that they are more influenced by personal traits than regional factors.

Implications:

The findings indicate the necessity for public health strategies that are spatially adaptive and account for localized socioeconomic and demographic disparities. In order to enhance crisis response effectiveness, policies targeting financial stability and access to health resources should be tailored to specific neighborhoods. Furthermore, this information can also provide a way for policymakers to design equitable interventions that better support vulnerable populations during health crises.

Future Work:

In the future, further research could refine spatial models by incorporating additional factors, such as mobility patterns and policy interventions. This would facilitate a more comprehensive understanding of regional decision-making processes. Expanding the dataset to include more regions and longitudinal data would allow for the tracking of behavioral changes over time. The incorporation of machine learning techniques could further improve predictive accuracy and reveal non-linear relationships between socioeconomic factors and decision-making. The integration of qualitative data could also provide deeper insights into the psychological and cultural influences that shape pandemic responses. These advancements would enhance public health strategies, ensuring more targeted and equitable crisis interventions.

VI. References

- [1] Gneezy A, Imas A, Jaroszewicz A. *The impact of agency on time and risk preferences*. Nat Commun. 2020;11:2665.
- [2] Ruggeri K, Panin A, Vdovic M, et al. *The globalizability of temporal discounting*. Nat Hum Behav. 2022;6:1386-1397.
- [3] Tversky A, Kahneman D. *The framing of decisions and the psychology of choice*. Science. 1981;211(4481):453-458.
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- [5] Fotheringham AS, Brunson C, Charlton M. *Geographically Weighted Regression: The Analysis of Spatially Varying Relationships*. Hoboken, NJ: Wiley; 2002.